Literature Snapshot v1

La(Fe,Si)$_{13}$ Magnetocaloric Data Layer

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Repository: `iliya-radulov/my_db_project` Branch: `main` Database: `alloy_lab` @ `192.168.178.10`

1 Purpose of this document

This report records the state of the La(Fe,Si)$_{13}$ literature database on 7 November 2026 (*Snapshot v1*). It serves three purposes:

- a reproducibility anchor: anyone can re-run the commands in Section 5 and obtain identical counts;
- a methods section for the Stage 4 Track B paper (the database design described in Section 3 is the first contribution of that paper);
- a decision log: data-source licences and the train/test separation are recorded in Section 6.

2 Snapshot numbers

Item	Count	Note
Records (`mce_literature_records`)	1 113	from 221 papers
Measurement points (`mce_literature_points`)	1 539	1 386 $\lvert\Delta s_T\rvert$ + 153 $\lvert\Delta T_{\mathrm{ad}}\rvert$
Distinct composition keys	660	129 keys appear in >1 paper (duplicates kept)
Lab samples with exact literature match	7	matched to 180 records
Sources currently loaded	1	`nims_lafesi_1_13`

Table 1: *Snapshot v1* counts, confirmed 7 Nov 2026.

3 Database schema

3.1 Overview

The literature data live in two tables (`mce_literature_records` and `mce_literature_points`) that are linked by a foreign key. A bridge table (`sample_composition_keys`) connects the literature records to the lab samples via a normalised composition key. Figure 1 shows the entity–relationship diagram.

3.2 Composition key

The composition key is the central join field. It is computed from the element amounts as follows:

1. Metal elements (La, Fe, Si, Mn, Cr, Ni, Co, ...) are normalised so their atom-percent sum equals 100 at.%.
2. Interstitial elements (H, C, N, B) are stored separately, appended after a `+` sign.
3. All amounts are rounded to 0.1 at.% resolution.

Example: `Fe81.4-La7.1-Si11.4+H3.6`. Two records are an exact match if and only if their composition keys are identical. This rule is intentionally strict: a composition reported as $La_1Fe_{11.4}Si_{1.6}$ in one paper and $La_1Fe_{11.3}Si_{1.7}$ in another yields two distinct keys and is kept as two separate records.

3.3 Entity–relationship diagram

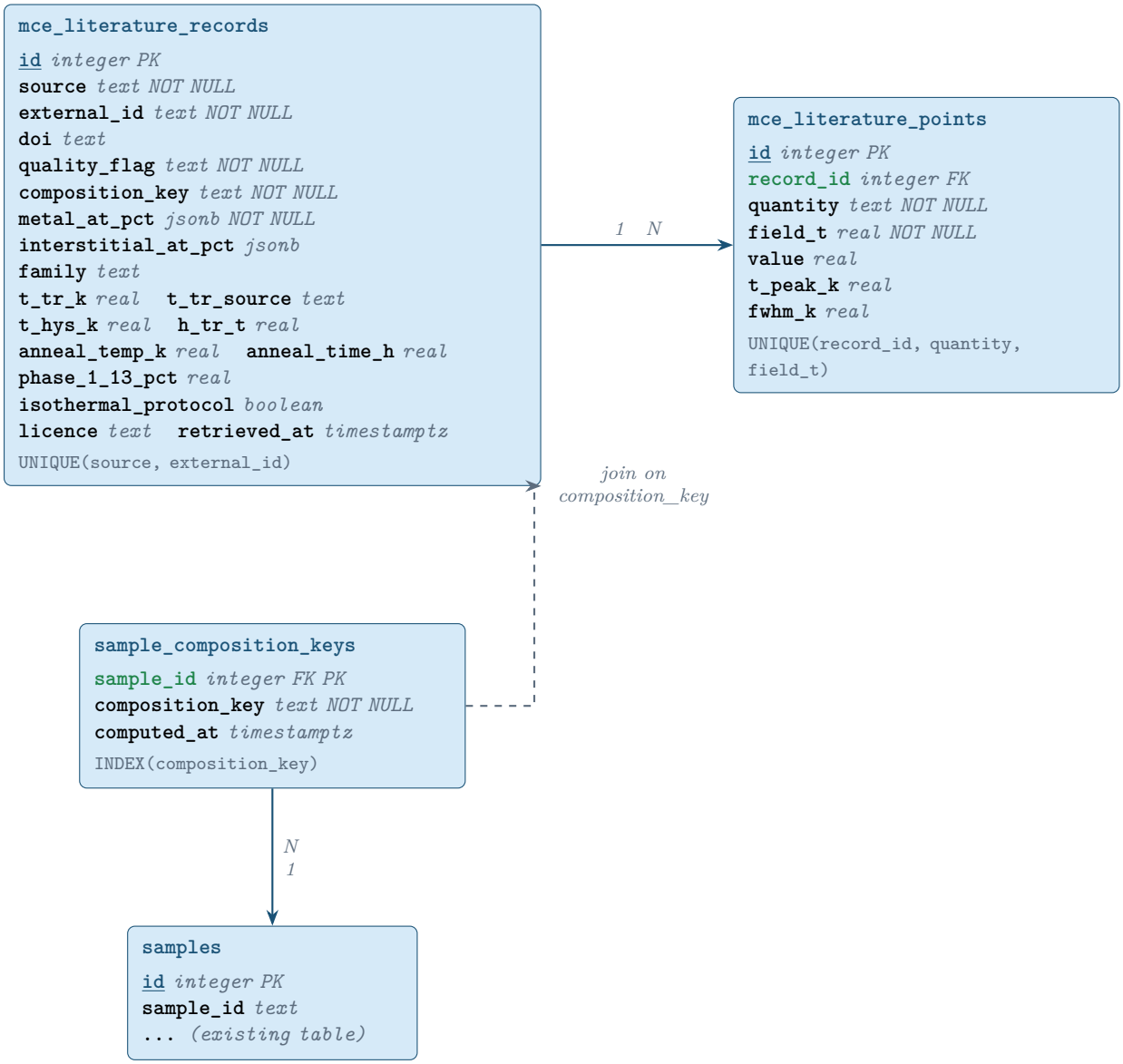


Figure 1: Entity–relationship diagram of the literature data layer. Solid arrows are foreign-key constraints; the dashed arrow is a logical join on `composition_key` (no FK constraint — a literature record may have no matching lab sample and vice versa). Primary keys are underlined; foreign keys are in green.

4 Import pipeline

Figure 2 shows the full flow from the raw NIMS file to the verified database state.

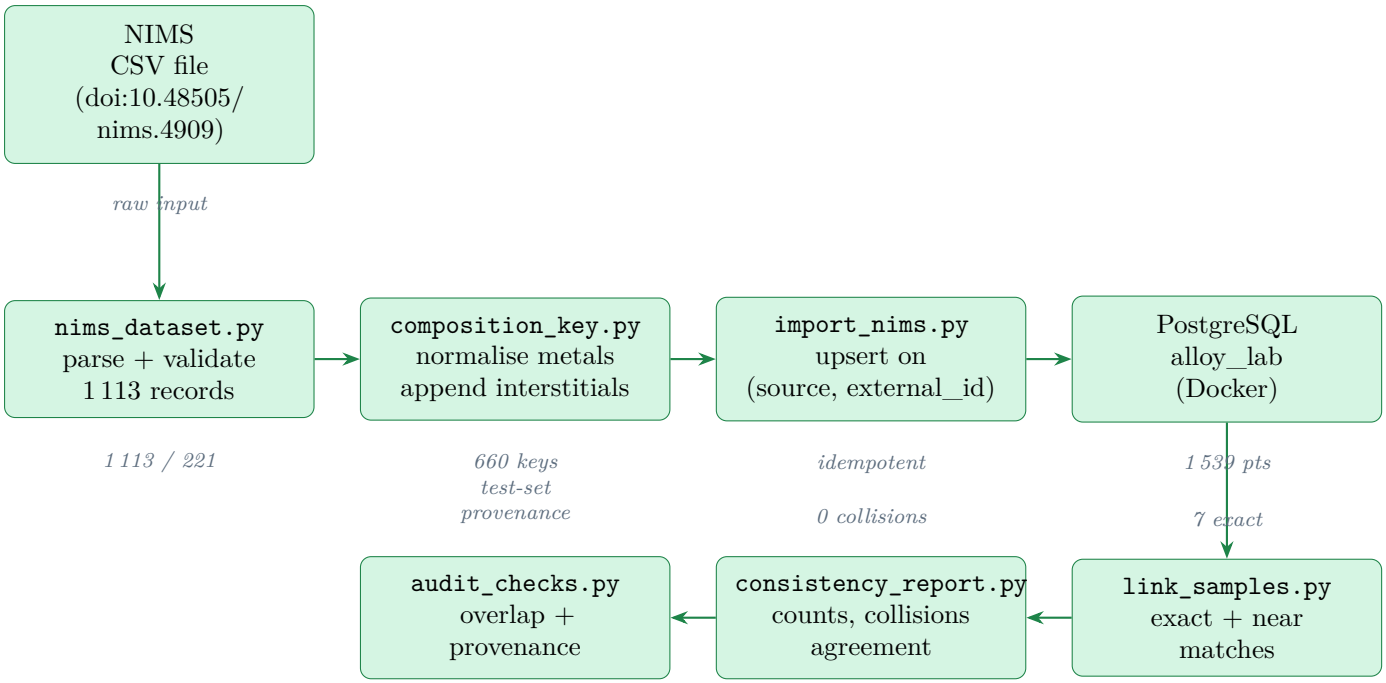


Figure 2: Import and verification pipeline for *Snapshot v1*. Every step is re-runnable; the upsert is idempotent.

5 Verification checks

All four checks were run on 7 November 2026 from the desktop machine (`192.168.178.47`) against the server (`192.168.178.10`).

The working directory was `~/Documents/github/active/my_db_project/stage_four/mce_nims/`.

5.1 Record and point counts

Listing 1: Database record counts.

```
$ ssh r@192.168.178.10 "docker exec alloy-postgres psql -U postgres \
    -d alloy_lab -c 'SELECT source, COUNT(*) as records \
    FROM alloy_lab.mce_literature_records GROUP BY source;'"

      source      | records
------------------+---------
 nims_lafesi_1_13 |    1113
(1 row)

$ ssh r@192.168.178.10 "docker exec alloy-postgres psql -U postgres \
    -d alloy_lab -c 'SELECT COUNT(*) as points \
    FROM alloy_lab.mce_literature_points;'"

 points
--------
   1539
(1 row)
```

5.2 Import idempotency (`import_nims.py -dry-run` and live run)

Listing 2: Dry-run and live import confirming idempotency.

```
$ python import_nims.py --dry-run
1113 records from 221 papers
  T_tr: {'reported': 835, 'dS_peak': 98, 'dS_dTad_mean': 26,
         'dTad_peak': 21}, missing 134
  dS_max points: 1386, dT_ad_max points: 153
  composition keys: 660 distinct; 129 keys appear in more than one paper
  quality_flag: 'experimental_literature' for all 1113 records
  dS: list lengths differ from the field list, values left empty:
      [(530, 'T_at_dS_max_K'), (531, 'T_at_dS_max_K')]

$ python import_nims.py
[... same summary lines ...]
Connected to database: alloy_lab
Imported 1113 records and 1539 points (source 'nims_lafesi_1_13')
```

The two list-length warnings on rows 530 and 531 are pre-existing entries in the NIMS source file (the temperature list has one fewer entry than the value list); the affected `T_at_dS_max_K` cells are left empty. This is not an import error.

5.3 Sample linking (`link_samples.py`)

Listing 3: Sample–literature linking output (abbreviated).

```
$ python link_samples.py
Connected to database: alloy_lab
73 samples: 44 keyed, 29 skipped
  skipped New_test_sample: not element symbols: ['placeholder']
  [... 28 further placeholder entries omitted ...]

Exact matches: 7 samples
  La1Fe113Si17-20260713-001  Fe80.7-La7.1-Si12.1   4 records from 4 papers
  La1Fe113Si17-20260713-002  Fe80.7-La7.1-Si12.1   4 records from 4 papers
  La1Fe114Si16-20260708-001  Fe81.4-La7.1-Si11.4  29 records from 21 papers
  La1Fe115Si15-20260708-001  Fe82.1-La7.1-Si10.7  40 records from 32 papers
  La1Fe115Si15-20260711-002  Fe82.1-La7.1-Si10.7  40 records from 32 papers
  La1Fe116Si14-20260708-001  Fe82.9-La7.1-Si10.0  50 records from 36 papers
  La1Fe118Si12-20260709-001  Fe84.3-La7.1-Si8.6   13 records from 8 papers

Near matches (every element within 0.5 at.%, excluding exact matches):
  La1Fe116Si14-20260708-001  Fe82.9-La7.1-Si10.0
    22 records; closest: nims_lafesi_1_13:174 (0.03 at.%)
  [...]
  La103Fe115Mn01Si14-20260902-001  Fe82.0-La7.3-Mn0.7-Si10.0
    3 records; closest: nims_lafesi_1_13:34 (0.20 at.%)
```

The 29 skipped samples all have `['placeholder']` as their composition entry — these are test or incomplete entries in the `samples` table and are expected to be absent from the matching.

5.4 Consistency report (`consistency_report.py`)

Listing 4: Consistency report output.

```
$ python consistency_report.py
Connected to database: alloy_lab

Per source
  source            quality_flag             records  papers  keys  with_t_tr
  nims_lafesi_1_13  experimental_literature  1113     221     660   979

Points per source
  source            quantity   points  records
  nims_lafesi_1_13  dS_max     1386    820
  nims_lafesi_1_13  dT_ad_max  153     128

New compositions (keys found in only this source)
  source            keys_only_here
  nims_lafesi_1_13  660

Collisions between sources
  keys_in_several_sources  papers_in_several_sources
  0                        0

Agreement where two sources have the same paper and composition key
  (none)

Own samples with exact matches
  source            quality_flag             samples  matched_records
  nims_lafesi_1_13  experimental_literature  7        180
```

Zero collisions are expected at *Snapshot v1*: the NIMS dataset is the only source loaded. The collision and agreement metrics become meaningful in Snapshot v2 once MCEdatabase and the Kelly & Cole datasets are imported.

5.5 Bracket-composition audit (section 4b of `SERVER_SETUP.md`)

This check identifies samples whose composition was entered with brackets or signs in the formula field, which an earlier version of the parser would have silently mis-parsed.

Listing 5: SQL audit for bracket-formula entries.

```
SET search_path TO alloy_lab;
SELECT sample_id, created_at, notes FROM samples
WHERE notes ~ 'Desktop App: .*[()\[\]{}+\-,;/]'
ORDER BY created_at;
```

Listing 6: Bracket audit result.

```
           sample_id            |         created_at         | notes
--------------------------------+----------------------------+---------------
 Fe195Co005P09Si01-20260714-001 | 2026-07-14 09:51:20.501258 |
   Added via Desktop App: pre-alloy Fe2P (10g)
   + raw Fe1.95Co0.05P0.9Si0.1 (at%)
(1 row)
```

One hit: a pre-alloy entry for a Fe_2P-family compound (not $La(Fe,Si)_{13}$). The note format `pre-alloy ... (g) + raw ... (at%)` is the expected pre-alloy pattern documented in `SERVER_SETUP.md` and requires no action.

6 Data sources and licences

Source	Status	Licence	Notes
NIMS $La(Fe,Si)_{13}$ doi:10.48505/nims.4909	**Imported**	CC BY 4.0	1 113 records / 221 papers; file not committed to git
MCEdatabase github.com/RunqingYang1996/MCEdatabase	**Permission confirmed**	Author email (Bolin Lau / Xuchao, 2 Nov 2026)	Reuse with citation of Yuan et al.; ≈202 records, 50 $La(Fe,Si)_{13}$ papers not in NIMS; not yet imported
Kelly & Cole 2025 figshare 29559686	Not yet imported	CC BY 4.0 (figshare)	≈100 records; direct contact not replied; figshare licence sufficient to proceed
Court & Cole 2018 figshare c.3954418	Not yet imported	CC BY 4.0 (figshare)	≈80 records; same as above
Dr. H. Ucar — Heusler dataset	**Received**	Personal comm.	≈500 Heusler records (Victorino 2018/19 + new); **future scope only** — not $La(Fe,Si)_{13}$

Table 2: Data sources at *Snapshot v1*. The NIMS file is kept only on the server and is not committed to the repository (CC BY 4.0 requires attribution but does not restrict private hosting).

Train / test separation (hard constraint)

The Moreno-Ramírez Ni and Cr papers are the held-out test set and are **never** used for training. This constraint is enforced in `evaluate.py` and must not be changed without a new snapshot document.

Training data decision

Adding MCEdatabase, Kelly & Cole, and Court & Cole to the NIMS training data showed no improvement in LOFO cross-validation MAE (pre-registered criterion, evaluated ≈4 November 2026). The decision: **train on NIMS only**. The additional sources are retained as provenance records and may be used as independent held-out validation in a later snapshot.

7 What comes next (Snapshot v2 plan)

1. Import MCEdatabase (`import_mcedb.py -dry-run` first).
2. Import Kelly & Cole from figshare (`import_curie_neel.py`).
3. Import Court & Cole from figshare (`import_court_cole_2018.py`).
4. Re-run `consistency_report.py`: expect non-zero collision and agreement rows.
5. One clean evaluation on the frozen Snapshot v2 dataset; record command lines and full MAE tables.

Reproducibility

Listing 7: Commands to reproduce *Snapshot v1* from scratch.

```
# 1. Import NIMS (idempotent)
python import_nims.py

# 2. Link lab samples
python link_samples.py

# 3. Consistency report
python consistency_report.py

# 4. Bracket-composition audit
ssh r@192.168.178.10 "docker exec alloy-postgres psql -U postgres \
  -d alloy_lab -c \"SET search_path TO alloy_lab; \
  SELECT sample_id, created_at, notes FROM samples \
  WHERE notes ~ 'Desktop App: .*[()\[\]{}+\-,;/]' \
  ORDER BY created_at;\""
```

Expected output: 1 113 records / 1 539 points / 7 exact sample matches / 0 collisions / 1 pre-alloy bracket entry.

Generated with Claude Sonnet 4.6. Database: PostgreSQL 17 in Docker (`alloy-postgres`). Scripts: `stage_four/mce_nims/` in `iliya-radulov/my_db_project`.